

# Lecture 10 Random Variables, PMF, Expectation & Variance

BIO210 Biostatistics

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## What is a random variable (r.v.) ?

- An assignment of a value (a real number) to every possible outcome in the sample space.
- **Mathematically:** A real-valued **function** defined on a sample space  $\Omega$ . In a particular experiment, a random variable (r.v.) would be some function that assigns a real number to each possible outcome.

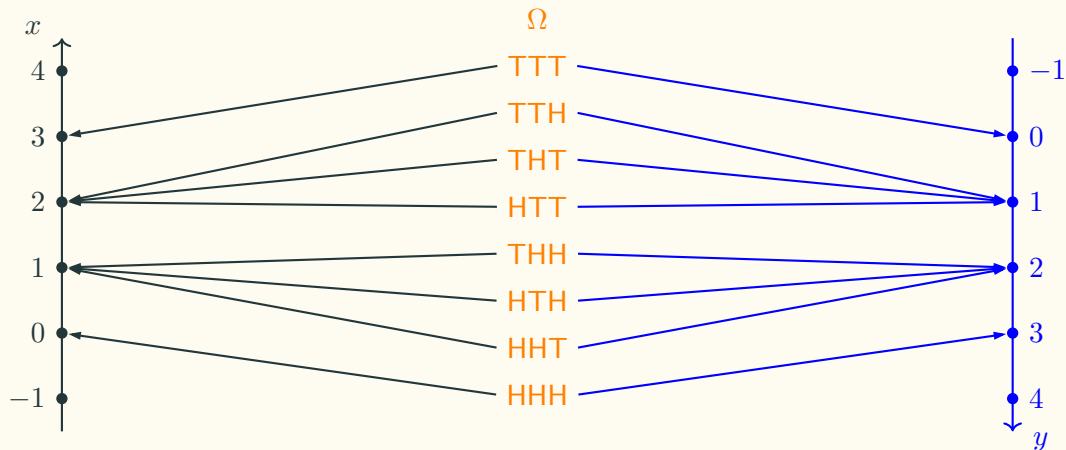
## More about random variables

- Discrete or continuous.
- Can have several random variables defined on the same sample space.
- Notation
  - random variable  $X$  : function  $\Omega \mapsto \mathbb{R}$
  - numerical value:  $x$  : value  $\in \mathbb{R}$

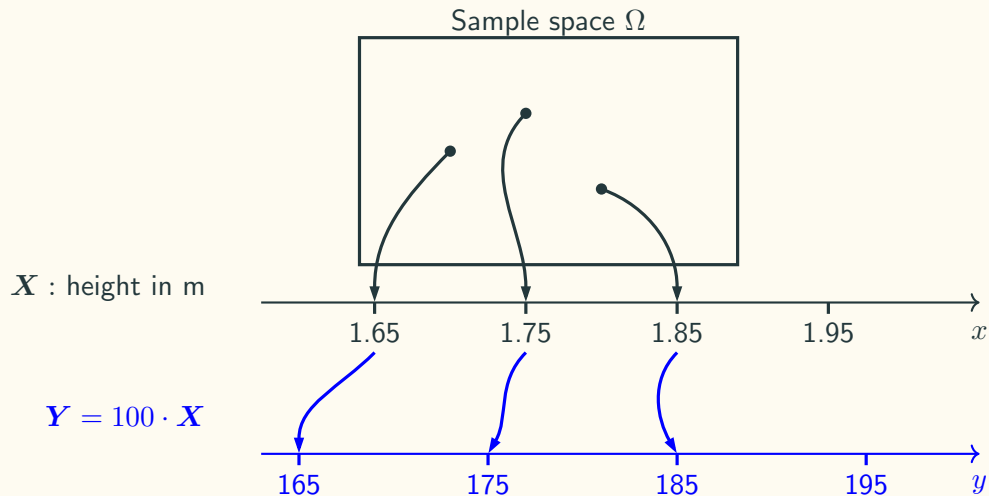
# Different random variables on the same sample space

$X$  : number of tails

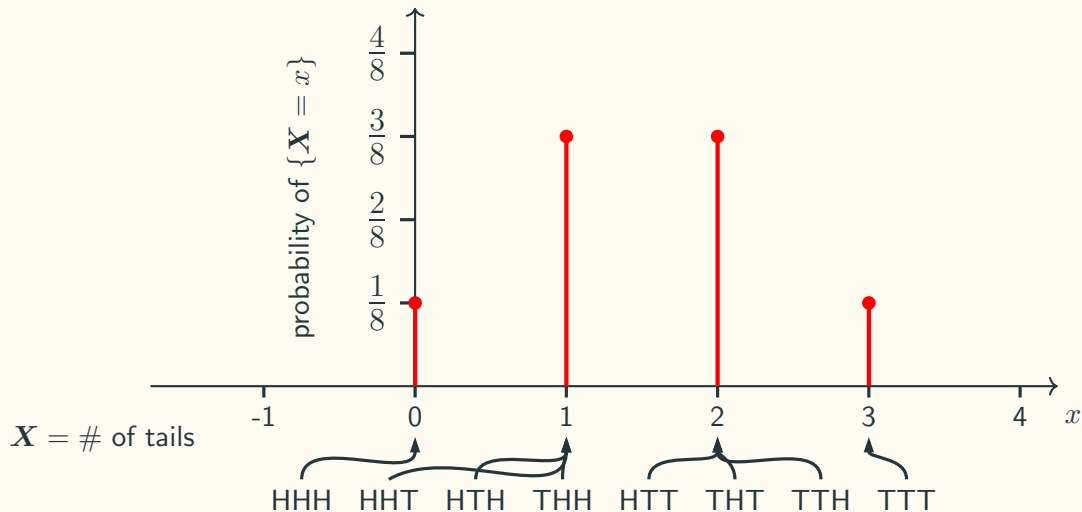
$Y$  : number of heads



Function of a random variable is an r.v.



# Probability Mass Function (PMF)



## Probability Mass Function (PMF)

The PMF of  $\mathbf{X}$  = number of tails after three flips

| $x$       | $\mathbb{P}(\{\mathbf{X} = x\})$ |
|-----------|----------------------------------|
| 0         | 1/8                              |
| 1         | 3/8                              |
| 2         | 3/8                              |
| 3         | 1/8                              |
| otherwise | 0                                |

$$\mathbb{P}(\{\mathbf{X} = x\}) = \begin{cases} \frac{1}{8}, & x = 0, 3 \\ \frac{3}{8}, & x = 1, 2 \\ 0, & \text{otherwise} \end{cases}$$

# PMF Notation

## Probability Mass Function

- Notation

$$\begin{aligned}\mathbb{P}_{\mathbf{X}}(x) &= \mathbb{P}(\{\mathbf{X} = x\}) \\ &= \mathbb{P}(\{\omega \in \Omega \mid \mathbf{X}(\omega) = x\})\end{aligned}$$

- Properties

$$\begin{aligned}\mathbb{P}_{\mathbf{X}}(x) &\geq 0 \\ \sum_x \mathbb{P}_{\mathbf{X}}(x) &= 1\end{aligned}$$

| $\omega$            | $\mathbf{X}(\omega) = x$ | $\mathbb{P}_{\mathbf{X}}(x) = \mathbb{P}(\{\mathbf{X} = x\})$ |
|---------------------|--------------------------|---------------------------------------------------------------|
| HHH                 | 0                        | $\frac{1}{8}$                                                 |
| TTH,<br>HTH,<br>HHT | 1                        | $\frac{3}{8}$                                                 |
| TTH,<br>THT,<br>TTH | 2                        | $\frac{3}{8}$                                                 |
| TTT                 | 3                        | $\frac{1}{8}$                                                 |

## Geometric PMF

**Experiment:** keep flipping a coin ( $\mathbb{P}(H) = p$ ) until a head comes up for the first time.  
Let the random variable  $X$  be the number of flips.

| $\omega$                            | $X(\omega)$ | $\mathbb{P}_X(x)$ |
|-------------------------------------|-------------|-------------------|
| H                                   | 1           | $p$               |
| TH                                  | 2           | $(1-p)p$          |
| TTH                                 | 3           | $(1-p)^2p$        |
| $\vdots$                            | $\vdots$    | $\vdots$          |
| $\underbrace{TTT \dots TTT}_{n-1}H$ | $n$         | $(1-p)^{n-1}p$    |

Geometric PMF.  $X$ : geometric random variable.

## How to compute a PMF $\mathbb{P}_X(x)$

**To compute a PMF  $\mathbb{P}_X(x)$ :**

1. Write all possible values ( $x$ ) that  $X$  can take;
2. For a value  $x$ , collect all possible outcomes for which  $X = x$ ;
3. add their probabilities;
4. repeat steps 2 & 3 for all  $x$ .

# Compute PMF

**Experiment:** two independent rolls of a fair tetrahedral die.

$F$ : outcome of the first roll

$S$ : outcome of the second roll

$$X = \min(F, S)$$

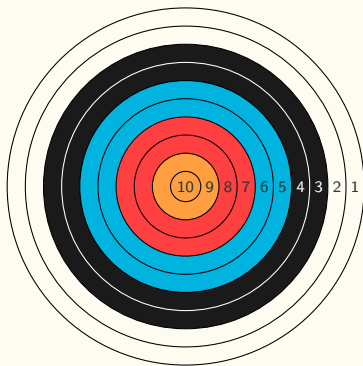
$$\mathbb{P}_X(x) = ?$$

|                   |   |                  |   |   |   |
|-------------------|---|------------------|---|---|---|
| $S$ : second roll | 4 |                  |   |   |   |
|                   | 3 |                  |   |   |   |
|                   | 2 |                  |   |   |   |
|                   | 1 |                  |   |   |   |
|                   |   | 1                | 2 | 3 | 4 |
|                   |   | $F$ : first roll |   |   |   |

# Expected value of a random variable (Expectation)

## Experiment: archery

Let  $X$  be the score you get for each shot. What is the expected value of  $X$  ?



| $x$ | $\mathbb{P}_X(x)$ |
|-----|-------------------|
| 1   | 0.19              |
| 2   | 0.17              |
| 3   | 0.15              |
| 4   | 0.13              |
| 5   | 0.11              |
| 6   | 0.09              |
| 7   | 0.07              |
| 8   | 0.05              |
| 9   | 0.03              |
| 10  | 0.01              |

Think: What is the average score you will get after a large number of trials?

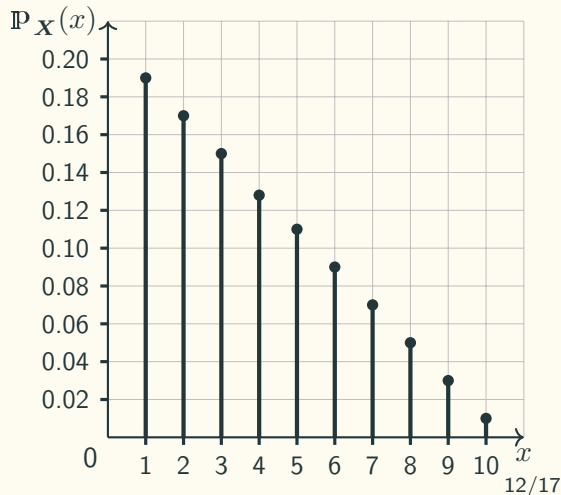
## Expected value (Expectation)

### Definition

$$\mathbb{E}[X] = \sum_x x \mathbb{P}_X(x)$$

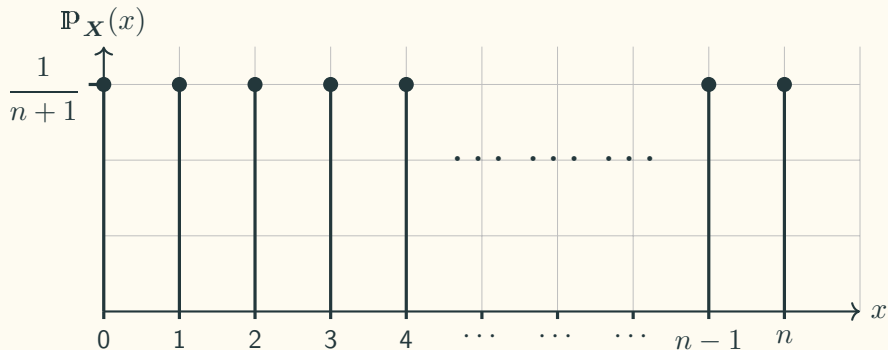
- Interpretation
  1. Centre of gravity of the PMF
  2. Average in large number of repetitions of the experiment

PMF of  $X$  from the archery experiment



# Expectation of a Uniform Distribution

**Example:** a uniform discrete random variable  $X$  on  $0, 1, 2, 3, \dots, n$



What is  $\mathbb{E}[X]$  ?

# Properties of expectations

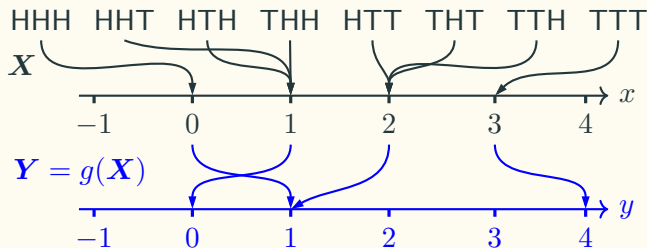
Let  $X$  be a random variable, and let  $Y = g(X)$ , what is  $\mathbb{E}[Y]$ ?

- The hard way:

$$\mathbb{E}[Y] = \sum_y y \mathbb{P}_Y(y)$$

- The easy way:

$$\mathbb{E}[Y] = \sum_x g(x) \mathbb{P}_X(x)$$



| $y$ | $\mathbb{P}_Y(y)$ |
|-----|-------------------|
| 0   | 3/8               |
| 1   | 4/8               |
| 4   | 1/8               |

| $x$ | $g(x)$ | $\mathbb{P}_X(x)$ |
|-----|--------|-------------------|
| 0   | 1      | 1/8               |
| 1   | 0      | 3/8               |
| 2   | 1      | 3/8               |
| 3   | 4      | 1/8               |

## Expectation of a linear function of r.v.

- Caution: in general  $\mathbb{E}[g(\mathbf{X})] \neq g(\mathbb{E}[\mathbf{X}])$
- Exception: if  $\alpha, \beta$  are constants, then we have:
  - $\mathbb{E}[\alpha] = \alpha$
  - $\mathbb{E}[\alpha \mathbf{X}] = \alpha \mathbb{E}[\mathbf{X}]$
  - $\mathbb{E}[\alpha \mathbf{X} + \beta] = \alpha \mathbb{E}[\mathbf{X}] + \beta$

# Variance and standard deviation of a random variable

## Definition of Variance

$$\mathbb{V}\text{ar}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}])^2]$$

## Properties of Variance

- $\mathbb{V}\text{ar}(\mathbf{X}) = \mathbb{E}[\mathbf{X}^2] - (\mathbb{E}[\mathbf{X}])^2$
- If  $\alpha, \beta$  are constants, then  $\mathbb{V}\text{ar}(\alpha\mathbf{X} + \beta) = \alpha^2\mathbb{V}\text{ar}(\mathbf{X})$

## Definition of Standard Deviation

$$\sigma_{\mathbf{X}} = \sqrt{\mathbb{V}\text{ar}(\mathbf{X})}$$

## Discrete Random Variables (Summary slide)

